

Thilakkumar V

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EDUCATION

National Institute of Technology, Tiruchirappalli

2020 – 2024

Bachelor of Technology in Civil Engineering, Minors in Computer Science

8.21 CGPA

Relevant Coursework: Data Structures, DBMS, Artificial Intelligence, Operating Systems, Software Engineering

Shree Niketan Matric. Hr Secondary School, Chennai

2020

Class XII, Central Board of Secondary Education

91.17%

EXPERIENCE

Product Engineer (Backend & DevOps) | Ganit Inc.

Aug 2024 – Present

An AI-first decision science company delivering enterprise-grade data solutions across industries

Chennai, TN

- Engineered AI-powered backend systems and real-time APIs using **FastAPI**, **Django**, and **WebSockets** for operational excellence in FMCG, HR tech, and healthcare domains.
- Built **7 real-time Sales Tower Dashboards** for an FMCG giant using **Node.js** and **Google BigQuery**; defined 20+ KPIs, authored 50+ optimized SQL queries for data insights, and improved operational performance by **40%** via automation and process optimization.
- Automated operational workflows through **CI/CD pipelines** for 4 production healthcare apps using GitHub Actions, Docker, and AWS, achieving **2x operational throughput** and reducing manual deployment errors by **80%**.
- Developed an **AI-powered Resume Parser Tool** using **Gemini API**, **LLMs**, and automation frameworks with Django REST; deployed via AWS, achieving **90% accuracy** and **75% reduction in manual effort** for operational tasks.
- Automated cloud infrastructure provisioning using **Terraform** and **AWS CloudFormation**; streamlined operations with **EC2**, **RDS**, **S3**, **VPC** automation, reducing deployment time by **60%** and operational overhead.
- Built real-time backend for **DatForge AI Chatbot** using **FastAPI**, **WebSockets**, and **AWS Bedrock GenAI**, achieving **sub-200ms latency**, **>99.9% uptime**, supporting **1000+ concurrent users** for operational efficiency.

Technical Product Analyst Intern | BlockTrack

May 2023 – Jul 2023

An IIT Madras incubated Health-tech startup specializing in Blockchain and SaaS products

Chennai, TN

- Conducted data-driven analysis on **50+ market players**, identifying trends and enabling tech partnerships.
- Leveraged **PEST analysis** to refine market positioning, boosting **profit margins by 5%** and **market share by 15%**.
- Designed and prototyped modular insurance workflows using BPMN tools and backend logic schemas, projected to enable a **75% expansion in user adoption**.
- Analyzed **2000+ structured survey responses** using Python (NumPy, Seaborn), generating actionable insights that informed **6 high-impact strategic initiatives**.

PROJECTS

NexEvent – Event Management Platform | React.js, Django REST, Docker, AWS, Analytics

Jun 2024 – Jul 2024

- Designed and deployed an **full-stack event management platform** supporting **5+ admin features** and **10K+ concurrent users**, with automated guest tracking, real-time attendance analytics, and predictive insights.
- Architected **RESTful APIs** with Django REST Framework to handle automated workflows, RSVP analytics, intelligent reminders, and data-driven operational insights.
- Built a responsive frontend using **React.js** and **Tailwind CSS** with real-time analytics dashboards, reducing page load time by **40%** and providing operational visibility across devices.
- Automated the end-to-end **operational pipeline** using Docker, GitHub Actions, and AWS CodePipeline; deployed to **ECS Fargate with autoscaling**, achieving **99.9% uptime** and operational excellence.
- Integrated **CloudWatch analytics** for real-time operational monitoring and automated alerting, reducing issue detection time by **70%** and enabling proactive operational management.

Predictive Drought Risk Model | Python, Machine Learning, Data Analytics, Automation

Jan 2024 – May 2024

- Integrated **historical climate data** & EC-Earth3 projections under CMIP6 framework, analyzing **6500+ rain gauge stations** for predictive analytics and operational insights.
- Enhanced **AI model accuracy by 20%** using advanced **Machine Learning techniques** (MLR, SVM) and automated bias correction for operational reliability.

- Automated **7.5 lakh analytical iterations** to quantify drought patterns, demonstrating expertise in large-scale data processing and operational automation.
- Designed **interactive analytics dashboards** and spatial visualizations using **Google Earth Engine**, providing actionable insights for operational decision-making.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL (MySQL, PostgreSQL), Bash, C/C++

AI/ML & Analytics: Machine Learning, Data Analysis, LLMs, Gemini API, AWS Bedrock, NumPy, Pandas, Seaborn

Frameworks: Django, FastAPI, ReactJS, Node.js, Express.js, Django REST Framework

Cloud/DevOps: AWS (EC2, S3, RDS, CloudWatch, Amplify, Bedrock), GCP, Docker, GitHub Actions, Terraform, CI/CD

Tools & Platforms: Git, Postman, VS Code, Linux, WebSockets, REST APIs, BPMN, Process Automation

Certifications: AWS Certified AI Practitioner

LEADERSHIP / ACTIVITIES

Operations Chairperson, Nittfest'23 | *The Annual Cultural Festival of NIT Trichy* **Mar 2022 – May 2024**

- Led **operational excellence** across a **3-tier team structure** of **250+ members**, orchestrating 80+ concurrent events for 12000+ participants through systematic process optimization.
- Managed comprehensive **operational budget of 75 lakh INR** with strategic resource allocation, cost optimization, marketing operations, and systematic risk management frameworks.
- Orchestrated complex **multi-stakeholder operations** featuring high-profile guests like **Dr. Kamal Haasan**, coordinating with **7+ production partners** through streamlined operational workflows.
- Achieved **70% increase in corporate partnerships** through **data-driven outreach optimization** and systematic process improvements across **50+ company collaborations**.

Consultant | *180 Degrees Consulting, NIT Trichy* **Aug 2022 – May 2024**

- Handled 4 consulting projects (3 startups, 1 NGO), with an **NPS of 100%** across clients.
- Within the **FinTech Innovation Project - BharatX**, conducted primary research on 10+ professional education loan systems, analyzed 5-year Indian Ed-Tech market growth with 7 parameters, and **devised 3 innovative strategies**.

ACHIEVEMENTS

- **Secured 2nd Position** at TransfINITe'23, a 42-hour long hackathon on software projects **Oct 2023**
- **Winner** - Bid and Build, eSummit'23, IIT Bombay **Feb 2023**